Media Bias ISM

RQ Methods Report

RQ1

Question. How has media reporting bias toward S&P 500 firms changed after the 2020 recession shock?

Variable Construction

For article a of firm i on date t, with MTSC probabilities (p_a^+, p_a^-, p_a^0), keep high-confidence observations only:

$$\max\{p_a^+, p_a^-, p_a^0\} \geq \tau, \quad \tau = 0.90 \text{ (default)}.$$

Define article stance component and firm-day outcome:

$$s_a = p_a^+ - p_a^-, \qquad \text{bias}_{it} = \frac{1}{|\mathcal{A}_{it}|} \sum_{a \in \mathcal{A}_{it}} s_a,$$

where $\mathcal{A}_{it}$ is the set of qualifying articles for firm i on day t.

Controls:

- $\text{vol}_{it} = |\mathcal{A}_{it}|$ (firm-day article volume)
- vix_t (daily market uncertainty)

Shock indicator:

$$\text{Post}_t = \mathbb{1}[t \geq 2020\text{-}01\text{-}01].$$

Baseline Model (Firm FE)

$$\text{bias}_{it} = \alpha_i + \beta\,\text{Post}_t + \gamma_1\,\text{vol}_{it} + \gamma_2\,\text{vix}_t + \varepsilon_{it}.$$

β is the main estimand: average within-firm post-2020 shift in stance.

Estimator and Inference

Within transform by firm means, then OLS:

$$\hat{\theta} = (\tilde{X}^\top \tilde{X})^{-1} \tilde{X}^\top \tilde{y}, \quad \theta = (\beta, \gamma_1, \gamma_2)^\top.$$

Robust HC1 covariance:

$$\widehat{\text{Var}}(\hat{\theta}) = \frac{n}{n-K} (\tilde{X}^\top \tilde{X})^{-1} \left(\sum_{j=1}^{n} \hat{u}_j^2 \tilde{x}_j \tilde{x}_j^\top \right) (\tilde{X}^\top \tilde{X})^{-1}.$$

Two-sided normal-approx p-values:

$$p_k = 2\,\Phi(-|t_k|), \quad t_k = \frac{\hat{\theta}_k}{\widehat{\text{SE}}(\hat{\theta}_k)}.$$

Identification Conditions

1. Conditional within-firm parallel trends after controls.
2. Controls are not post-treatment bad controls in the identification sense.
3. **Stable measurement process:** NewsMTSC score mapping is comparable over time.

Results: Confidence-Threshold Sensitivity

RQ1 was re-estimated three times with confidence thresholds $\tau \in \{0.0, 0.6, 0.9\}$. For each experiment, the table reports FE coefficients, article counts, and within-R^2.

Table 1: *
Table RQ1-A: FE Results at Confidence Threshold $\tau = 0.0$

Variable	Coef.	SE	t-stat	p-value	CI Low	CI High
Post	0.0208	0.0033	6.3568	0.000000	0.0144	0.0272
Article Volume	-0.0006	0.0001	-7.8722	0.000000	-0.0008	-0.0005
VIX	0.0002	0.0002	0.7053	0.4806	-0.0003	0.0006
Raw Articles (after filters)	462,788					
Firm-day Observations (n)	49,136					
Within R^2	0.001792					

Table 2: *
Table RQ1-B: FE Results at Confidence Threshold $\tau = 0.6$

Variable	Coef.	SE	t-stat	p-value	CI Low	CI High
Post	0.0232	0.0036	6.4321	0.000000	0.0161	0.0302
Article Volume	-0.0009	0.0001	-8.4881	0.000000	-0.0011	-0.0007
VIX	0.0002	0.0002	0.6800	0.4965	-0.0003	0.0006
Raw Articles (after filters)	378,573					
Firm-day Observations (n)	46,681					
Within R^2	0.002041					

Table 3: *
Table RQ1-C: FE Results at Confidence Threshold $\tau = 0.9$

Variable	Coef.	SE	t-stat	p-value	CI Low	CI High
Post	0.0258	0.0052	4.9425	0.000001	0.0155	0.0360
Article Volume	-0.0028	0.0003	-9.4603	0.000000	-0.0033	-0.0022
VIX	0.0005	0.0004	1.4886	0.1366	-0.0002	0.0012
Raw Articles (after filters)	136,787					
Firm-day Observations (n)	31,575					
Within R^2	0.002613					

Interpretation. Across thresholds, the post coefficient remains positive and statistically significant. As the threshold increases, article coverage declines while the post effect magnitude increases slightly and within-R^2 rises modestly.

RQ2

Question. Did the 2020 recession shock affect stock prices differently pre versus post, at the market level?

Design Rationale

RQ2 is paired with RQ1. RQ1 tests whether media stance shifts after the shock, while RQ2 tests whether firm-level returns also shift. Both are required before claiming a structural media–market linkage in RQ3.

Variable Construction

For firm i and date t, daily return is computed from close prices:

$$\text{return}_{it} = \frac{P_{it} - P_{i,t-1}}{P_{i,t-1}}.$$

Shock indicator:

$$\text{Post}_t = \mathbb{1}[t \geq \text{2020-01-01}].$$

Controls:

- sp500_return_t: absorbs market-wide bull-run and broad index drift.
- vix_t: absorbs volatility-regime shifts common to all firms.

Baseline Model (Firm FE)

$$\text{return}_{it} = \alpha_i + \beta\,\text{Post}_t + \gamma_1\,\text{sp500\_return}_t + \gamma_2\,\text{vix}_t + \varepsilon_{it}.$$

As in RQ1, estimation uses entity fixed effects and HC1 robust standard errors.

Results

Sample and diagnostics. The estimation panel contains 62,667 firm-day observations across 26 firms (2015-12-01 to 2025-11-24). Pre-period means are tightly clustered (range = 0.00193, SD = 0.00042).

Table 4: *

Table RQ2-A: Entity FE Regression (HC1 Robust SE)

Variable	Coef.	SE	t-stat	p-value	CI Low	CI High
Post	-0.000344	0.000172	-2.0038	0.0451	-0.000681	-0.000008
S&P 500 Return	1.127576	0.010982	102.6767	0.0000	1.106052	1.149100
VIX	-0.000001	0.000020	-0.0433	0.9654	-0.000039	0.000038

Interpretation. Controlling for index return and VIX, the post-period shift in firm daily returns is negative and marginally significant (about -0.034% per day). The market-return control is strongly positive and highly significant, while VIX is not statistically significant in this specification.